DNA Replication

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Video animation

Main video for DNA replication:

https://youtu.be/CRw05N1_R5Q

Additional video:

https://youtu.be/WCliyI4__cM

Key words from the video

1. **DNA Replication:** The process of making an identical copy of DNA.
2. **Leading Strand:** The DNA strand replicated continuously in the 5' to 3' direction.
3. **Lagging Strand:** The DNA strand replicated discontinuously in the 3' to 5' direction.
4. **DNA Polymerase:** An enzyme that synthesizes a new DNA strand using a template strand and adds nucleotides to the 3' end of a growing polynucleotide chain.
5. **Semi discontinuous:** Refers to the combination of continuous and discontinuous replication on the leading and lagging strands, respectively.
6. **Helicase:** An enzyme that unwinds the DNA double helix during replication.
7. **Replication Fork:** The point where the two DNA strands are separated during replication.
8. **Primase:** An enzyme that synthesizes short RNA primers on each DNA template strand.
9. **RNA Primer:** Short sequences of RNA that provide a starting point for DNA synthesis.
10. **Okazaki Fragments:** Short DNA fragments formed on the lagging strand during discontinuous replication.
11. **DNA Ligase:** An enzyme that connects Okazaki fragments by forming bonds between adjacent DNA nucleotides.

Key words from the video

1. **Nick:** A gap or break in the DNA strand where the sugar-phosphate backbone is incomplete.
2. **Chromosome:** A thread-like structure composed of DNA and proteins that carries genetic information.
3. **Continuous DNA Synthesis:** The uninterrupted synthesis of a new DNA strand on the leading strand template.
4. **Discontinuous DNA Synthesis:** The synthesis of a new DNA strand on the lagging strand template in fragments.
5. **Template Strand:** The original DNA strand used as a guide to synthesize a new complementary strand.
6. **3' to 5' Direction:** The direction in which DNA synthesis cannot occur, but necessary for the lagging strand synthesis.
7. **5' to 3' Direction:** The direction in which DNA synthesis proceeds during replication.
8. **Replication Termination:** The point where DNA replication concludes, either at the end of the chromosome or when another replication fork approaches from the opposite direction.

Why is DNA replication considered semi discontinuous?

a. Both strands are replicated continuously.
b. One strand is replicated continuously, and the other is replicated discontinuously.
c. Both strands are replicated discontinuously.
d. Neither strand is replicated.

What is the role of helicase during DNA replication?

a. Synthesizing RNA primers
b. Unwinding the DNA double helix
c. Connecting Okazaki fragments
d. Recognizing and removing RNA primers

What is the function of primase in DNA replication?

a. Synthesizing DNA nucleotides
b. Unwinding the DNA double helix
c. Creating RNA primers
d. Connecting Okazaki fragments

Why does DNA polymerase need RNA primers during replication?

a. To unwind the DNA double helix
b. To provide a starting point for DNA synthesis
c. To connect Okazaki fragments
d. To recognize and remove RNA primers

What are Okazaki fragments, and where are they found?

a. Long DNA strands on the leading strand template
b. Short RNA primers on the lagging strand template
c. Short DNA fragments on the lagging strand template
d. RNA molecules on the leading strand template

What is the role of DNA ligase in DNA replication?

a. Synthesizing RNA primers
b. Connecting Okazaki fragments
c. Unwinding the DNA double helix
d. Synthesizing new DNA strands

What is the consequence if DNA ligase does not eliminate the "nick" in the DNA strand?

a. DNA replication would continue indefinitely.
b. The DNA strand would remain incomplete.
c. Okazaki fragments would not form.
d. Helicase would reattach to the DNA double helix.

What happens during continuous DNA synthesis on the leading strand template?

a) DNA polymerase moves away from the replication fork.
b) DNA ligase forms bonds between adjacent DNA nucleotides.
c) Helicase unwinds the DNA double helix.
d) A new DNA strand is synthesized without interruption.